

TITLE PAGE: (Child Development):

2- and 3-Year-Olds' Coordinate Decisions in Dyads and Triads

Authors: Rivero, Alissa.¹, Vasil, Jared.², and Michael, Tomasello.¹

Affiliations:

1. Department of Psychology & Neuroscience, Duke University¹, Durham, NC
2. Department of Psychology, New York University², New York, New York

Address: 2200 W Main St., Durham, NC 27705

Telephone: 240-863-4505

Fax: 919.660.5726

Email Address: alissa.rivero@duke.edu

Conflict of Interests: None

Funding: Jacob's Foundation

Data Availability statement: All data and analyses are made publicly available at the following OSF Link:

https://osf.io/qjbk5/overview?view_only=b40d34212cbc49ef9bdee5a0111fdc4b

Author Contributions Statement:

Alissa Rivero: Conceptualization, data curation, formal analysis, investigation, methodology, writing (original draft and review & editing)

Jared Vasil: Conceptualization, methodology, formal analysis, visualization, writing (original draft and review & editing)

Mike Tomasello: Conceptualization, funding acquisition, methodology, project administration, resources, supervision, writing (review & editing)

Abstract

Even young toddlers coordinate actions with a partner. But coordinating choices or decisions is more challenging, particularly in multi-party groups. In this pre-registered study (data collected from Fall 2023 to Spring 2025), 2- and 3-year-olds ($n = 100$, 50% female, 26% underrepresented minority) needed to select “keys” that matched in color to open a box. Children coordinated with one another in same-aged dyads or triads. In dyads, children of both ages were mostly successful, whereas 3-year-olds were much more successful than 2-year-olds in triads. This was especially true in a generalization trial in which only one of ten 2-year-old triads was successful, while all other groups were successful. Success at both ages was predicted by rates of cooperative communication about the keys and their color. These results demonstrate that even 2-year-olds can coordinate choices with a single partner, but doing so in groups tends to emerge at around age three, when children begin to develop more advanced ways of thinking about and working with others.

Keywords: coordination, cooperation, group

2- and 3-Year-Olds Coordinate Decisions in Dyads and Triads

Coordinating actions with others may occur through simple reactive and anticipatory processes, such as the way flocks of birds travel together in mesmerizing patterns, or the way human drivers effortlessly avoid one another in traffic. But humans, unlike any other species, coordinate not just actions, but *decisions*, with others on small and large scales, made possible by cooperative communication (Tomasello, ____). Yet, it is unclear when this capacity for coordinating decisions with others in groups emerges.

From soon after their first birthday, young children engage in simple collaborative activities with adults (Warneken et al., 2006, 2007). But at this early age, they struggle to collaborate with peers. In free play, there is little true collaboration among toddlers (Howes & Matheson, 1992). Instead, what appears to be cooperation can often be explained by the relatively simple process of one child following the other to some toy and doing the same thing (leader-follower pattern; Eckerman et al., 1989). When encouraged to collaborate in experiments, children are first able to coordinate their actions with a peer only after 2 years of age (Brownell & Carriger, 1990). Further, compared to 18-month-olds, 2-year-olds made more active attempts to coordinate their actions with their peer partner, for example, by communicating or moving around to different sides of the apparatus. Presumably, children collaborate earlier and more easily with adults than with peers because adults compensate for children's limitations by considering the child's perspective, anticipating what information they will need, and communicating plans. For these same reasons, toddlers also engage in both joint attentional interactions (Ninio, 2016) and linguistic conversations (Eckerman & Didow, 1996; Eckerman & Peterman, 2001) more skillfully with adults than with peers. Still,

humans go beyond the basic collaboration and coordination of actions observed in 2-year-olds.

In game theory, coordination games involve more than just coordinating actions; they involve coordinating decisions or choices. For example, imagine that two children on the playground are each independently told that their friend wants to play with them inside the school building. But, they don't know in which of several rooms they should meet. Without communication or oversight of the other child's decision, this would be difficult. The difficulty is that without communication, we must take into consideration where we expect the other person to go, in addition to where they expect us to go, leading to a never-ending loop of recursive inferences (Schelling, 1960). If the children could communicate before deciding where to go – and they knew to do so – meeting each other would of course be much easier.

So far, only children of four years of age and older have been tested in such coordination games. In a game of so-called pure coordination, Duguid et al. (2020) presented pairs of 4-year-olds with a very straightforward task: partners were situated on either side of an apparatus containing four options, and they had to choose the same option to make resources available for both (they learned this in pretraining). The majority of 4-year-olds were successful, most often by communicating their choice to their partner (via either pointing or language) before selecting it. Interestingly, in this same study, great apes were sometimes successful, but their success resulted mostly from chance coordination followed by them continuing to choose the same option trial after trial. They simply learned that a particular option was the rewarded one, further evidenced by the fact that when paired with a new partner (with different experience), they failed. On the other hand, children were just as

competent with a new partner, indicating that they understood that success depended not on a particular choice but on coordinating one's choice with one's partner.

In a Stag Hunt coordination game, pairs of 4-year-olds were in the process of accessing low-value resources (the “hare” box) when a high-value resource (the “stag” box) suddenly became available (Duguid et al., 2014). But the rewards in the stag box were accessible only if partners worked on it together (which they each knew from pretraining), and so the choice of the stag was somewhat risky: it would only lead to success if both partners chose it. The pairs were mostly successful in deciding to leave their low-value options and coordinate on the higher-value stag, again most often by communicating pre-decision. And again, great apes approached the task very differently: they did not communicate or coordinate pre-decision. Rather, one individual went for the stag straight away, and the other simply followed (leader-follower strategy).

In more difficult coordination tasks, success has been demonstrated only in still older children. For example, in an experiment in which children needed to choose a complementary tool by anticipating and planning for their partner's tool choice, 3-year-olds did not consistently make the right choice, whereas 5-year-olds were much more successful (Warneken et al., 2014). As might be expected, 3-year-olds who communicated were more likely to succeed than those who did not. Finally, the most difficult coordination problems are those in which there is no possibility of communication (e.g., Grüneisen et al., 2015a, 2015b, 2017). These have been solved only by 5-year-old children, who engage in a kind of recursive mind-reading (each choosing what they expect the other to expect them to choose, etc.) using salient “focal points”. For instance, the children in our initial example might choose the room in which they played yesterday, or the one with a special toy.

It is unknown whether children younger than 4 years of age can successfully coordinate their choices and decisions in coordination games. If communication is possible, we might expect 2-year-olds to be at least somewhat competent, since they successfully coordinate actions and have begun developing basic skills of communicating information with and seeking information from peers (Brownell & Carriger, 1990). However, knowing how to coordinate choices and decisions proactively by communicating relevant information pre-decision may still be challenging for children this young, due to cognitive limitations in such things as executive function, working memory, planning, and perspective-taking. Based on these developing mechanisms and other theoretical considerations, 3-year-olds should be significantly more competent than 2-year-olds.

A further question is whether children's earliest coordinative decisions with peers are limited to dyadic interactions, or whether the underlying mechanisms also support collective decision making in larger groups. Research in social psychology has revealed a host of differences between interacting dyadically with a single partner and interacting in more complex and impersonal group settings (Moreland, 2010). The basic problem is that simply adding one more partner to constitute a triad exponentially increases the difficulty involved in aligning decisions. Thus, to extend our hypothetical scenario, if three children were involved, the likelihood of coordinating on a single room becomes much less likely. However, again, with communication, coordinating a decision with two others, or even a larger group, should be relatively straightforward.

In the current study, we observed young children from 2;0 to 4;0 years of age in a novel pure coordination task. Each child had to choose the same color key as their partner(s) to open a locked box and retrieve a token reward. The placement of the keys and box were

such that children could not see which one their partner(s) chose, and thus, to ensure reliable and efficient coordination, children needed to communicate their decision or move around the table to see which key their partner(s) chose. After three identical test trials, children were given (without any training or instruction) a generalization trial in which they had to match not colors but pictures on the keys, with successful generalization indicating a clear understanding of the need to coordinate. We observed children in same-aged dyads and triads, predicting that while there might be gains in success from two to three years of age in dyads, there would be a more dramatic difference in success from two to three years of age in triads, both because operating in triads is more cognitively and socially complex and because research has shown that around three years of age, children begin to understand much more about operating in groups (Tomasello, 2019, 2024).

Methods

Participants

The final sample included 100 children (50% female, 26% underrepresented minority), for a total of 40 groups with 10 groups per condition at each age. Thus, ten 2-year-old dyads, ten 2-year-old triads, ten 3-year-old dyads, and ten 3-year-old triads were included in the final sample. The 2-year-olds ranged from 2;0 to 3;0 (mean = 2.52), and the 3-year-olds ranged from 3;0 to 4;0 (mean = 3.47). Sixteen additional groups were excluded based on failure to complete all trials (nine 2-year-old triads, four 2-year-old dyads, two 3-year-old triads, and one 3-year-old dyad). The exclusion rate was largely driven by the challenges involved in testing multiple 2-year-olds at once. Therefore, included participants were potentially more willing to engage with their peers and the task than excluded participants.

We recruited children from Duke University’s child development participant database or through their local daycares in Durham, North Carolina between Fall 2023 and Spring 2025. Ten percent of the 2-year-old groups were tested in their daycare, compared to forty percent of the 3-year-old groups, and this distribution was driven by the number of available children at each age at local daycares. All other children were tested in a laboratory setting. This study was approved by the IRB at Duke University and preregistered through AsPredicted (<https://aspredicted.org/6fz4-nsmt.pdf>).

Design

We employed a 2 x 2 between-subjects design, with age (2-year-old; 3-year-old) and group size (dyad; triad) as factors. Children were randomly assigned to a group-size condition (within some practicality constraints), without consideration of gender or any other demographic factors. Children completed a basic pure coordination task (Figure 1). Regardless of age or group size, the task was the same for the children: put the *same* key on the box to unlock it and gain access to the rewards inside.

In terms of the apparatus, two or three children sat around a square table. In the dyad condition, there were two chairs placed across from each other, and in the triad condition, three chairs were arranged around three sides of the table. In the center of the table, a wooden box was velcroed down to prevent children from moving or twisting it. The box had a glass lid, allowing children to see the reward tokens inside (called “coins”). There were always two “coins” per collaborator. Children learned that they could drop the coins in a music box to make a fun noise and were motivated to get more coins. On the table, between each child and the box, were two magnets: one red and one blue (for the children, they were called “keys”).

Each child could select a key and stick it on their side of the box, such that children could not see each other's chosen color when they were attached to the box or on the table. Children were informed that the box would unlock when they placed matching color keys (the unlocking mechanism was actually controlled wirelessly by experimenters). The music box was placed approximately four feet from the table.

Procedure

Parents or legal guardians provided informed consent, and children provided verbal assent before participation. Children played with each other for a few minutes in a warm-up area to grow more comfortable with the experimenters, the testing space, and their partner(s).

Before attempting the coordination task in their dyad or triad, children were individually trained on how the box worked with an adult experimenter. In the triad condition, the child was either trained with two experimenters (when tested at their daycare) or the experimenter and their caregiver (when tested in the lab). First, each child was introduced to the music box and given reward tokens (coins) to demonstrate how the box could emit music. Then, the experimenter(s) brought each child over to the square table (with nothing on it yet), describing how they would learn how to get more coins together. Then, the experimenter distributed one red and one blue key to each collaborator. In the dyad condition, children sat across from one experimenter, while children in the triad condition sat at the table with both experimenters. Children were then told that before getting more coins, they needed to learn how to match. The experimenter taught the children how to match keys to criterion, meaning each child correctly matched the red and blue keys in two consecutive turns.

After this training on how to match based on color, the experimenter introduced the

wooden box, pointing to the coins through the glass lid while saying, “*See, look, there are more coins!*” The experimenter demonstrated that the box was locked, stating, “*Uh Oh! The box is locked. To open the box, we both/all need to place the same color.*” Children practiced opening the box three times, with the experimenter(s) prompting the use of different colors to match in each attempt. Furthermore, the experimenter(s) attempted to encourage behaviors relevant to coordination (e.g., communicating, pointing, showing) during the individual training. While children were not trained to criterion in this phase of the individual training, each participant was able to open the box with adult scaffolding.

Following the sequential individual training of each child, all children participating in the dyad/triad were brought into the testing area at the same time. They were asked to sit at the table where the keys, but not the wooden box, were laid out for each participant. Once all children were seated, the experimenter announced that the box was coming and reminded the children to pick the same color. For this practice trial (the group warm-up), the box was not initially placed on the table to allow children time to decide on a color before placing a key on the box. This group warm-up was exactly the same, no matter whether children participated in a lab or school setting. The experimenter told each child to “*Pick your color*” and then reminded the group, “*Remember, you all need to pick the same color.*” Ten seconds later, the experimenter placed the box on the table, pointing out the coins inside, telling the children it was “*Time to start!*” Every 10 seconds, children were given increasingly informative prompts in a set order (regardless of children’s progress in the task) until the box opened. The prompts were as follows: (1) “*Remember, you need to work together!*”, (2) “*Remember, you need to pick the same color.*”, (3) “*You don’t have the same one yet!*”, (4) “*_____ has red/blue(, _____ has red/blue,) and _____ has red/blue on the box.*” If needed, prompts 3 and 4 were

repeated, resulting in a total trial length of 70 seconds (unless children opened the box sooner). If children failed to open the box within the 70-second trial, the experimenter opened the box for them. Children were rarely successful toward the end of the trial (Supplementary Material Figure 1), so we do not believe that the 70-second cutoff placed artificial constraints on children's success opening the box.

Following the practice trial, children participated in three test trials with each other. The test trials were nearly identical to the practice trial, with the only difference being that the box started on the table. As in the practice trial, participants received informative prompts at regular intervals. Following those three test trials, children participated in a final generalization trial that featured a change of key type. In this final trial, children were given new keys and told, "*Now, instead of red or blue, you have pig and cow.*" Children were given no additional instruction to see if they would generalize their knowledge of matching colors to matching animals. In the generalization trial, prompts 2 and 4 were modified to "*Remember, you need to pick the same animal*" and "*_____ has pig/cow...*" In all test trials, experimenters stood back from the task and avoided eye contact, and children's attempts to communicate with the experimenter(s) about the task were ignored.

Coding and Reliability

Children's behavior was coded by the first author. A second and third coder, both unaware of the hypotheses, coded 20% of the video recordings each, for a total reliability subset of 40%. The preregistered coding scheme was adjusted due to (i) the low frequency of certain variables (e.g., normative language), (ii) an inability to code some variables from the tapes due to poor sight lines, and (iii) an oversight in the preregistration: our omission of

strategy. Final measures were coded according to the procedures described below:

Individual Engagement: For each trial, each child's behavior was coded as: (1) off task, meaning they did not actively try to open the box at any time during the trial, (2) on task, meaning they tried to open the box at some point during the trial, or (3) on task and social, meaning they were tried to open the box and interacted with their peers at some time during the trial. Percent agreement on individual engagement was 84.4% (weighted $\kappa = .71$).

Success: Whether or not groups succeeded in opening the box in each trial (binary). Percent agreement on success was 100% ($\kappa = 1$).

Key: Which key (red or blue, or pig or cow) successful groups used to open the box in each trial. Percent agreement on key used was 100% ($\kappa = 1$).

Strategy: If groups successfully opened the box, their method of doing so was coded into two categories at the group level: individualistic and collaborative. Individualistic strategies included (i) chance, if they happened to immediately land on the same key, (ii) trial and error, if they opened the box by switching between their keys until the box opened or switched their key in response to an experimenter prompt, (iii) simple strategy, where dyads/triads repeatedly used the same key (i.e. always picked blue) to open the box, and (iv) supplanting, if one child placed a key on the box for at least one of their collaborators. Collaborative strategies included (i) looking at their partner(s)' choice to see what key to use, or (ii) communicating which key to use (either verbally or nonverbally by pointing or showing). Percent agreement on strategy was 87.3% ($\kappa = .81$).

Communication: The number of communicative acts in each trial that were directed at a peer and in reference to a specific key (summed across children; agreement was 88.1%), divided by the number of prompts (a proxy for time), agreement was 96.9%; weighted $\kappa = 1$).

Both successful and unsuccessful groups could be assigned a value of 7 prompts, the maximum number of prompts received. Communication was coded only from the start to end of each trial (i.e., when the box opened or 70 seconds passed, whichever came first).

Analysis Plan

The data was analyzed with Bayesian generalized linear mixed models via brms (Bürkner, 2017) in R (Version 4.5.2). Bayesian inference quantifies the credibility of hypotheses by defining probabilities based on prior beliefs and collected data (Gelman et al., 2013). We used Bayesian inference for two reasons. First, our use of weakly to moderately informative priors aided our models' convergence while discouraging large estimates unless warranted by the data. Second, Bayesian regression allows each model to answer a broader range of questions than an analogous frequentist model. For example, tests against chance, any contrast of interest (including those with recoded parameters), marginal predictions, probable direction of an effect, etc., can all be interpreted from one model.

Our priors were informative in the sense that they assumed that parameter values were normally distributed with means centered on chance or zero (depending on the parameter) with a specified variance. Our use of normal priors centered on zero informed the model that both directional hypotheses were equally likely prior to seeing the data. In other words, our use of priors was mainly for purposes of regularization (Gelman et al., 2017), rather than to bias the model toward our directional prediction (Vanpaemel, 2012). Further, by slightly concentrating mass around zero, our priors weakly pulled our parameter estimates towards zero, unless the data strongly indicated otherwise. Prior variances were initially determined based on a combination of domain expertise (e.g., plausible ranges of effect sizes on the logit

scale) coupled with sensitivity and reliability testing (i.e., the model was iteratively refitted and modified to ensure consistently robust convergence).

We investigated our directional hypotheses by computing the posterior probability that the predicted mean is greater, rather than less, than a comparison value. These posterior probabilities represent the probability that a predicted mean or effect takes the hypothesized direction, given the model, data, and averaging performed. In the absence of relevant visuals, we report the triplet of posterior quantities [lower bound of the 95% highest density interval (HDI), posterior mean, upper bound of the 95% HDI]. When describing posterior quantities, the triplet values are on the probability scale, range from 0 to 1, and chance is .5. When describing differences between posterior quantities (i.e., effects), triplet values may be negative, and chance is 0.

The supplement contains details about the analysis plan, inference criteria, sampler characteristics, convergence diagnostics, and posterior predictive checks (all adequate per Gelman et al., 2013), model formulas, families, link functions, priors, and model posterior parameter estimates. Random effects were always included when warranted by the operationalizations of the dependent variables (whether they were coded at the individual-level versus group-level, or trial level versus session-level). Code and data to reproduce all analyses, data, and results figures are available at the following OSF link (https://osf.io/qjbk5/overview?view_only=e6a4ab47e1204ff0a909138949ff8af8).

Results

Confirmatory Analyses

Children were generally engaged with the task, being completely off task in just 1.3%

of trials. They actively engaged with both the task and their peers in 60.3% of trials and individually with the task in 38.5% of trials. Engagement rates did not vary by participant age or group size (see additional reporting on this “engagement” model in the supplement). The 3-year-olds were not simply more engaged with the task than 2-year-olds, suggesting that engagement was not a low-level explanation for the condition differences. The remaining analyses investigate children’s success, communication, and cooperative strategies.

How successful were dyads and triads at the pure coordination task?

We hypothesized that children’s success at coordinating decisions with more than one partner would undergo a shift at around 3 years of age; specifically, that 2-year-old triads would succeed at low rates relative to 2-year-old dyads and 3-year-old dyads and triads. Consistent with our hypothesis, 2-year-old triads were uniquely unsuccessful: their success rate was roughly half that of the other groups (Table 1). Overall, with the exception of the 2-year-old triads, children’s performance was near ceiling in this simple pure coordination task. Moreover, three 2-year-old triads never succeeded on the task, while every other group succeeded at least once. There was no evidence of learning across trials, even in the 2-year-old triads who had room to improve.

In the final generalization trial, where children needed to match animals instead of colors, 2-year-old triads were once again uniquely unsuccessful, this time much more dramatically so (Table 2). Only one out of ten 2-year-old triads was successful, whereas 100% of the 3-year-olds and 2-year-old dyads succeeded.

To analyze children’s success rates across conditions, we fit a hierarchical logistic regression model, predicting success across all four trials from group size (dyad or triad),

group age (the mean age of individuals in the group), their interaction, testing location (school or laboratory setting), and familiarity. While familiarity and location were correlated, there were children tested in the lab who knew each other (e.g., twins and friends), and there may be effects of operating in a familiar environment (not just with familiar peers); thus, they were modeled as separate predictors. There were random intercepts of group ID and trial (see additional reporting on this “coordination” model in the supplement). As there were no learning effects across trials, trial was not included as a fixed effect. While older groups were, on average, more successful than younger groups, this was primarily driven by the triads. In other words, consistent with our hypothesis, there was a strong interaction between age and group size: triads improved more with age than dyads (Figure 2), posterior probability = 1.00, [0.132, 0.276, 0.425]. There were no directional effects of location or familiarity on participants’ predicted success rates: location: posterior probability = .75 that children succeeded more often in school than in the lab, [-0.068, 0.040, 0.162]; familiarity: posterior probability = .59 that unfamiliar peers communicated more than familiar peers [-0.096, 0.014, 0.129].

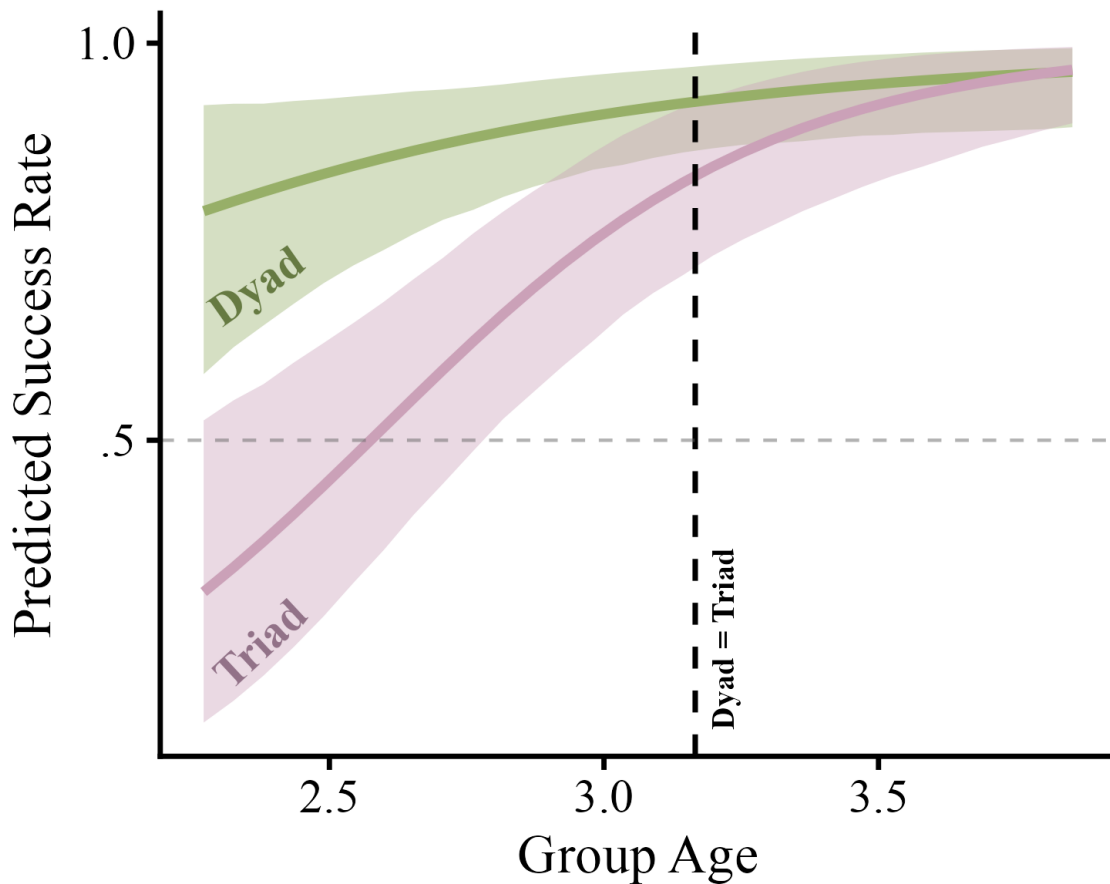


Figure 2. Predicted Success Rates. Solid lines represent predicted success rates across all four trials by group age and group size (posterior means). Group age is the mean age of group members in years. Bands represent middle 95% posterior intervals. The y -axis represents the predicted probability of successfully opening the box on a given trial, with a probability of 0.5 representing chance (denoted by the horizontal broken line). The vertical broken line represents the youngest age at which dyads and triads succeeded at nondifferent rates (details below and in Supplementary Material Figure 2).

Using the same posterior model, we conducted a post-hoc analysis—a Bayesian analogue of a Johnson-Neyman test—to determine at what age children were as successful in this task in triads as they were in dyads. Children began to succeed at nondifferent rates in dyads and triads by 3.17 years of age (Figure 2; details in Supplementary Material Figure 2). It is important to note that children’s success rates at 3.17 are remarkably high in all conditions; thus, the Johnson-Neyman does not necessarily suggest that 3.17-year-old

children are equally competent at coordinating in dyads and triads, but that children in both dyads and triads reach near-ceiling performance in this simple coordination task by 3.17 years of age.

Altogether, the pattern of children's successes aligned with our predictions. Children's success in triads gained faster with age than did their success in dyads. Further, by 3.17 years of age, children succeeded at similar rates in triads and dyads, indicating that the basic capacity to coordinate decisions in triads emerges shortly after the third birthday.

How much did children communicate?

Given the task of matching keys, the communication most relevant to coordination involves children communicating which key to use to their partner(s) before the box opened. Thus, this is how we operationalized cooperative communication. Regarding children's overall communication (i.e., any task-relevant communication), 3-year-old groups communicated a total of 181 times across 80 trials, while 2-year-old groups communicated a total of 163 times across 80 trials. However, of children's total communication, 75.0% of 3-year-olds' communication was in reference to a specific key, compared to just 54.2% of 2-year-olds' communication. Similarly, of children's overall communication, 74.0% of 3-year-olds' communication was directed toward a peer, compared to just 44.8% of 2-year-olds' communication. Thus, 3-year-old children's communication during the task was slightly more frequent overall, and more likely to be relevant to coordination: there were 127 instances in which a 3-year-old communicated about a specific key to peers, compared to just 70 instances of the same in 2-year-olds.

We analyzed whether older children communicated more than younger children, whether children communicated more in triads than dyads, whether successful groups communicated more than unsuccessful groups, and whether children in schools communicated more than children in laboratory settings. To this end, we modeled communication as a function of group age, group size, and success (successful or unsuccessful), the two-way interactions of each of those predictors, location, and familiarity. In this hierarchical Poisson regression model, there were random intercepts of group and trial, and a random slope of outcome within group. To model groups' rates of communication per prompt (to standardize communication across varying trial lengths), an offset term was used (see additional reporting on this "communication" model in the supplement).

Regarding communication relevant to coordination specifically, 3-year-olds communicated approximately 2.5 times more often than 2-year-olds (0.84 communicative acts per prompt interval, compared to 0.34 communicative acts, respectively), and this directional effect of age was statistically reliable, posterior probability = 1.00 that 3-year-olds communicated more often than 2-year-olds, [0.103, 0.270, 0.442]. Moreover, successful groups talked approximately 4.5 times more often than unsuccessful groups (0.71 communicative acts per prompt interval, compared to 0.15 communicative acts, respectively), and this directional effect was supported by the model; posterior probability = .97 that successful groups communicated more often than unsuccessful groups, [-0.013, 0.140, 0.291]. Dyads and triads communicated at similar rates (0.66 communicative acts per prompt interval, compared to 0.52 communicative acts, respectively), and this was reflected in similar estimated communication rates, posterior probability = .20 that triads communicated more often than dyads, [-0.231, -0.068, 0.087]. There was no interaction between group size and

group age.

Children tested at school (1.60 communicative acts per prompt) tended to talk more than did children tested in the lab (0.85 communicative acts per prompt), posterior probability = .91, [-0.060, 0.111, 0.289]. Similarly, there was a tendency for familiar peers (1.41 communicative acts per prompt) to communicate more often than unfamiliar peers (0.84 communicative acts per prompt), posterior probability = .91, [-0.045, 0.103, 0.271]. Overall, regarding communication relevant to coordination, 3-year-olds communicated more frequently than did 2-year-olds, successful groups communicated more frequently than unsuccessful groups, school groups communicated more frequently than lab groups, and familiar peers communicated more frequently than unfamiliar peers.

Exploratory analysis of children's communication relevant to coordination reveals another interesting difference between 2- and 3-year-olds, with potential implications for children's success. Children could communicate about a specific key to their peer(s) very simply, for example by stating a color name or pointing to a key. Indeed, this was the most frequent form of children's cooperative communication across both age groups, with 40 of these communicative acts in 2-year-olds and 45 in 3-year-olds. Yet, while these basic communicative acts comprised 75.5% of 2-year-olds' cooperative communication, they accounted for only 43.0% of 3-year-olds' cooperative communication. Three-year-olds often communicated in more sophisticated ways than did 2-year-olds: 30.8% of their communicative acts described which key they selected or requested info about which key to select (e.g., "I picked blue" or "did you pick pig?"), 15.0% were attempts to correct their partner(s)' incorrect color (e.g., "No! You don't have the same color!"), and 11.2% were directives or suggestives (e.g., "Put the cow one on."). Altogether, children's communication

became more sophisticated and relevant to coordination with age, potentially explaining why older children were more successful at the task.

Did cooperative communication predict success?

We investigated whether communication predicted success by fitting another hierarchical logistic regression model. In this model, we predicted success by the interaction of communication and group size and included random intercepts of group and trial (see additional reporting on this “coordination-communication” model in the supplement). The communication predictor tracked the number of communicative references to a specific key (or keys) directed towards peers, per individual per prompt. We expected that groups’ chances of successfully opening the box would increase as they talked more. Indeed, this was supported by our posterior model (Figure 3), as dyads’ and triads’ predicted success rates increased the more that they talked to each other about the keys: dyads: posterior probability = .99, [0.011, 0.042, 0.084]; triads: posterior probability = 1.00, [0.081, 0.133, 0.186]). While both dyads and triads benefited from communication, the effect was stronger in triads, posterior probability = .99, [0.035, 0.091, 0.152]. Altogether, these results suggest that both dyads’ and triads’ success was aided by their cooperative communication, but to a greater extent in triads.

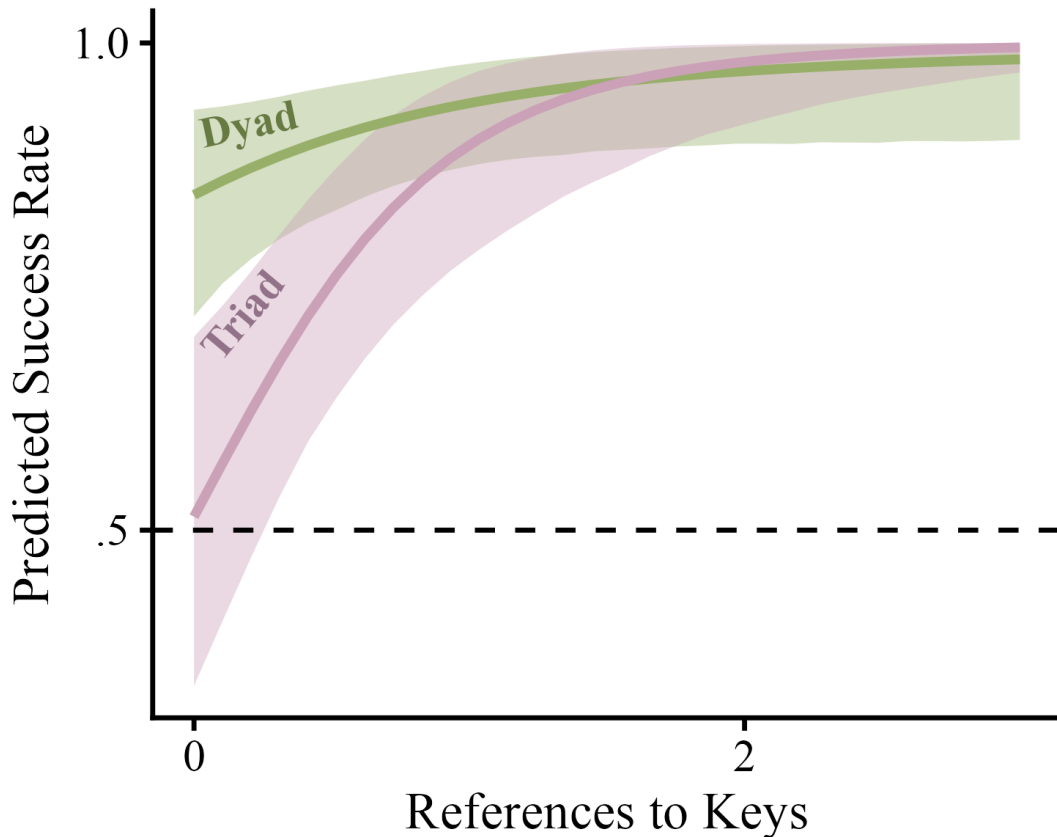


Figure 3. Communication Predicts Success Rates. Success rate predicted by groups' communication. Solid line represents posterior mean. Bands represent middle 95% posterior intervals.

Exploratory Analyses

Still, an important question remains: *why* did 2-year-olds succeed most of the time in dyads, but less than half of the time in triads, even though the task itself was identical (put the same key on the box)? To answer this question, we investigate whether children's success reflected coordinated decision-making, or whether their apparent coordination resulted from lower-level individual behaviors.

How did successful groups coordinate?

The previous model of success (Figure 3) predicts that dyads who do not communicate at all still perform near ceiling, while triads who do not communicate succeed only about half of the time, suggesting that dyads effectively compensated in other ways. Indeed, children could successfully match keys in the pure coordination task without coordinating at all. Children could have matched colors by chance, through trial and error, by placing their partner(s) key(s) on the box for them, or by choosing the same color every time. We describe these methods as individualistic strategies, as success did not *require* children to consider their partner(s)' choice. In contrast, collaborative strategies by definition involved children actively monitoring their partner(s)' selected key or communicating their choice to their partner(s). Three-year-old groups, on average, succeeded more often via collaborative strategies (Table 3).

We analyzed whether and in which conditions successful groups tended to succeed via collaborative strategies as opposed to individualistic strategies. In this logistic regression model, we predicted the type of strategy used by successful groups (collaborative or individualistic) from the interaction of communication and group size, with random intercepts of group and trial (see additional reporting on this “strategy” model in the supplement). This analysis consisted of data from 37 groups, rather than our entire sample of 40, because three groups (all 2-year-old triads) never succeeded once. The 3-year-olds succeeded via collaborative strategies more often than individualistic strategies, posterior probability = .99, [.543, .659, .791]. In contrast, 2-year-olds did not clearly favor either strategy, posterior probability = .16, [.259, .415, .591]. Thus, when they succeeded, 3-year-olds were more likely than 2-year-olds to have done so by collaborating, posterior probability = 1.00, [0.078, 0.244, 0.398]. Altogether, these results suggest that 3-year-olds were more likely to succeed by

cooperatively aligning their plans about which key to use with partners, whereas 2-year-olds tended to succeed via more individualistic strategies like chance or trial and error.

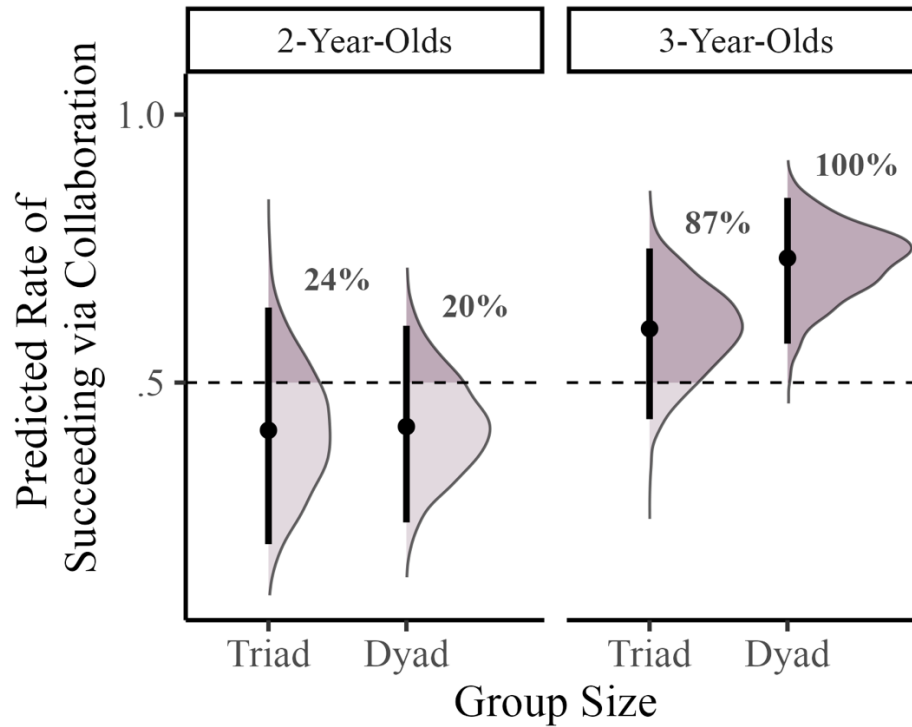


Figure 4. Coordination Strategies. Values above .5 (i.e., chance, denoted by a broken horizontal line) indicate a tendency to use a collaborative strategy (dark purple), and values below a tendency to use an individualistic strategy (light purple). Percentages indicate relative evidence for the statistical hypothesis that collaborative strategies were used more often than individualistic strategies.

Of the individualistic strategies, choosing the same color across each of the three test trials is of particular theoretical interest. Given that continuing to choose the first successful color can be explained by basic reinforcement learning principles, and that apes tend to employ the same rule learning strategy (Duguid et al., 2020), we examined color selection strategies across the three test trials (Figure 5). For groups who were successful in at least two out of three test trials ($n = 33$), their color selection strategy was operationalized as “first

successful” (i.e., use same color magnet to open the box in each successful test trial), “alternate” (i.e., alternate between colors in each successful test trial), or “no pattern” (i.e., no pattern in color selection). This data was used to fit a categorical model comparing observed rates to chance. Details about the computation of chance ($\sim .29$) are available in the supplement, as well as additional reporting on this “color” model.

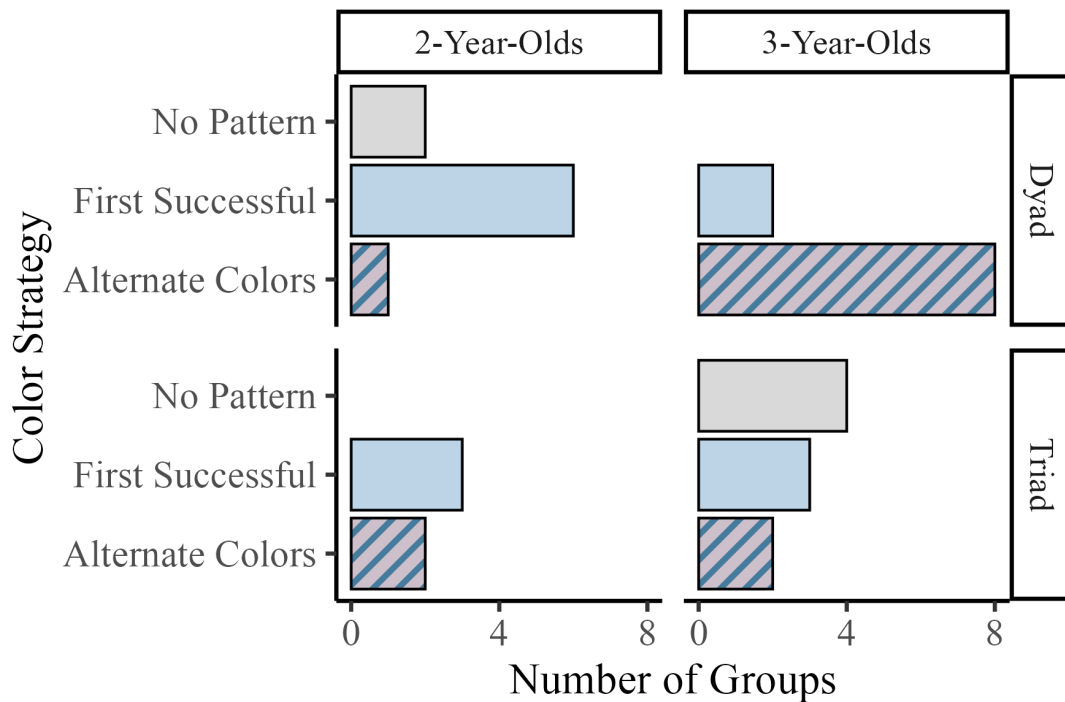


Figure 5. Key Color Selection Strategies. The x-axis represents the number of successful groups who chose keys according to the strategy depicted on the y-axis.

Successful 2- and 3-year-old groups relied on categorically different color coordination strategies (full posteriors depicted in Supplementary Material Figure 3). Two-year-olds tended to continue using their first successful color across both dyads and triads at rates above chance: dyads: posterior probability = .98, [.303, .555, .793], triads: posterior probability = .94, [.240, .534, .822]. We urge caution in interpreting the former model prediction for 2-year-old triads, which we think is overconfident relative to what the data

offer. Specifically, because 2-year-old triads were observed infrequently in this analysis ($n = 5$), the model partially pooled its estimate for them towards its estimate for the 2-year-old dyads ($n = 9$), who were observed nearly twice as often and had a similar empirical rate of first successful strategy use (.66 among dyads and .60 among triads). Importantly, though, when collapsing 2-year-olds' color selection strategies across group sizes, 2-year-olds' success can be largely explained by picking the same color across trials.

In contrast to 2-year-olds, 3-year-olds did not show the same preference for first successful strategy use, indicating that they were not simply relying on a learned rule. Three-year-old triads did not show a preference for one color pattern over the other (all directional effects probabilities were .56 to .68). Remarkably, 3-year-old dyads showed a robust preference for a different color pattern by alternating between red and blue more often than chance, posterior probability = .99, [.379, .591, .807]. Alternating colors demonstrates a sophisticated understanding of coordination – that it is not the color but the collective agreement that matters.

Altogether, 3-year-olds' successes were largely predicated on their efforts to coordinate decisions. On the other hand, 2-year-olds' success can mostly be explained by individual efforts that masquerade as coordinated behavior due to proximity and emergent phenomena (selecting the same key). Importantly, 2-year-olds' individualistic strategies were sufficient in dyads: choosing the same key by chance or by trial and error is far more likely to work when trying to match with one partner than two. However, the individualistic strategies were not sufficient to reliably overcome the challenges involved in coordinating in triads, explaining why 2-year-old triads were uniquely unsuccessful. Notably, in the most stringent test of children's coordination, the generalization trial in which children could not rely on

choosing the previously successful color, 2-year-old triads struggled mightily. The collaborative strategies used by the 3-year-olds are more robust and flexible, explaining why 3-year-olds were remarkably successful both in triads and in the generalization trial.

Discussion

The 3-year-old children in the current study clearly understood the coordination problem and were skillful at solving it in both dyads and triads. In contrast, 2-year-olds in the current study were successful most of the time in dyads, but less than half the time in triads. There were several other indications that the 3-year-olds were more skillful at coordinating. Whereas 2-year-old dyads mostly used more individualistic strategies, such as picking the same color each trial, 3-year-old dyads mostly succeeded by collaborating, and they communicated in ways relevant to coordination more often. Nevertheless, despite these age differences in strategy, in the generalization trial at the end – when the colored keys were changed out for keys with pictures of animals – all dyads at both ages were successful. The simple, individualistic strategies employed by 2-year-olds were sufficient in dyads. On the other hand, only one out of ten 2-year-old triads was successful, compared to all 3-year-old triads; when 2-year-old triads could no longer rely on choosing a previously successful color, but needed to find a new basis of coordination, they suddenly were not competent. The age of transition for successful coordination in triads – that is, the age at which children were as successful in triads as in dyads – was 3.17 years.

It is important that all children in all conditions received extensive training to criterion with adults (either in dyads or in triads, depending on condition) before testing. Before testing, all children in both conditions successfully demonstrated the ability to match colors to

open the box and to communicate their color choice to partner(s). Thus, while children of both ages were able to fully participate in both dyads and triads when scaffolded by adults, when this scaffolding was removed, 2-year-old children regressed, especially in triads and particularly in the generalization trial. This regression occurred even despite a practice trial with their peer partner(s) in which they were given the opportunity to communicate and agree on a key before they could act on the box (as a way of encouraging communication),

Communication was clearly crucial to the success of children of both ages in both conditions. Children of both ages communicated at similar rates, but 3-year-olds' communication was far more relevant to task success. Most important, regardless of age and condition, children's rate of task-relevant communicative acts predicted success: the more children talked about a specific key to their partner(s), the greater their probability of success. Importantly, while both dyads and triads benefited from such communication, the effect was stronger in triads (presumably because the individualistic strategies were less helpful in triads). Importantly, the communication skills required for consistent success in this task (telegraphic speech) were well within the competence of even the youngest children, and indeed all children showed relevant communicative competence in the pretraining with adults. It would thus seem that it was not linguistic skill or frequency per se that accounted for age differences in communication in this task, but rather children's understanding that communication was important for coordination.

In an immediate sense, then, the explanation for the developmental pattern of children's performance in this task was their understanding of the importance of coordinating communication – and the inadequacy of individual strategies – for sustained task success. But since coordination and communication are so closely bound together in the task, one might

also ask if there are underlying cognitive or social competencies that explain the observed age trends in both of these, especially in the triadic condition. Cognitively, because coordinating in triads is more complex than coordinating in dyads, one obvious candidate is children's developing skills of executive function/working memory, which develop rapidly between two and six years of age (Doebel, 2020). From two to four years of age, children improve on many complex tasks, often in ways that correlate with their emerging skills of executive function/working memory, which enable them to hold more things in mind at one time. This could account for the age trends in both the dyadic and triadic conditions.

Socially, children in this age range are rapidly developing new skills as well. In a naturalistic task, 3-year-olds demonstrated an understanding of their (adult) partner's false belief, suggesting greater theory of mind and perspective-taking ability (Köymen et al., 2026). Additionally, experimental studies of children's social cognition show an important transition at age three, specifically in their understanding of beliefs, social conventions, language, pedagogy, social commitments, fairness, and social norms (see Tomasello, 2019, Table 11.1). Arguably, the new level of understanding in all these cases involves 3-year-olds' emerging understanding of functioning in cultural groups – as distinct from functioning directly in face-to-face dyads (Moreland, 2010) – and their artifacts and products. If children are indeed developing a “group-mindedness” at around their third birthdays, this would account especially for their growing skills in coordinating with multiple partners in triads: the older children were better able to think about the coordinated actions and success of the group as a whole, whereas the younger children were more focused on individual actions. There may also be life history explanations that account for 3-year-olds' competence engaging with peers. In traditional societies, age three is the age of weaning when children begin to be left

on their own more often to make their own way with peers (Hrdy, 2006), and naturalistic studies of children's peer interactions show growing competence during this age range (e.g., Rubin et al., 2006).

The most likely hypothesis, we believe, is that both cognitive and social factors were at work in our study. Future research to investigate these different possible underlying cognitive and social mechanisms could take two tacks. First, investigators could measure such things as executive function/working memory, social cognition, and group-mindedness in children who also participate in coordination tasks to see if there is a correlation over this age range. Second, investigators could engage in training tasks of one type or another to try to boost 2-year-old children's performance. This training could involve elements of the task itself (for example, more training in coordination and/or cooperative communication) or it could involve the hypothesized underlying mechanisms of executive function, skills of peer interaction and/or “theory of mind”, or more specifically group-mindedness. Finally, another extremely interesting extension of this research would be testing children from more small-scale traditional societies, who spend more time at younger ages operating in peer groups than children from WEIRD societies (Lew-Levy & Amir, 2026).

The current study was limited in several ways. First, recruiting triads of children (especially 2-year-olds, many of whom are not yet in preschool) was difficult. Although we ended up with 100 children total, we were only able to have as a final sample ten groups per experimental cell. Relatedly, our exclusion rate was relatively high ($n = 16$), driven primarily by challenges involved in testing multiple 2-year-olds at once (our 2-year-old participants may thus have been especially skillful in social interaction in general). Second, we tested an unequal proportion of children in schools versus the lab across age and group size (due to

difficulties both in assembling triads in the lab and in accessing preschools with enough children of the right ages). Children tested in schools communicated more frequently than those tested in the lab, and more of the 3-year-olds came from schools, and so our age differences must be interpreted with caution. One possible causal factor is that children in their schools were interacting with children whom they knew at least to some degree (which was mostly not true in the lab), and so future research should investigate this variable – that is, specifically, familiarity – more systematically.

In all, the current study has demonstrated that children as young as two and three years of age can succeed in solving pure coordination problems – the youngest age demonstrated to date. Further, we provide evidence that 3-year-olds not only coordinate actions, but also decisions (often through cooperative communication), allowing them to coordinate in groups and flexibly in new situations. Future research should seek to investigate more fully the cognitive and social mechanisms that underlie this advance in group-level coordination around the third birthday.

\

References

- Brownell, C. A., & Carriger, M. S. (1990). Changes in cooperation and self-other differentiation during the second year. *Child Development*, 61(4), 1164–1174. <https://doi.org/10.2307/1130884>
- Bürkner, Paul-Christian. (2017). brms: An R Package for Bayesian Multilevel Models Using Stan. *Journal of Statistical Software*. 80. 10.18637/jss.v080.i01.
- Doebel S. (2020). Rethinking Executive Function and Its Development. *Perspectives on Psychological Science: A Journal of the Association for Psychological Science*, 15(4),

942–956. <https://doi.org/10.1177/1745691620904771>

Duguid, S., Wyman, E., Bullinger, A. F., Herfurth-Majstorovic, K., & Tomasello, M. (2014).

Coordination strategies of chimpanzees and human children in a Stag Hunt game.

Proceedings of the Royal Society B: Biological Sciences, 281(1797), 20141973.

<https://doi.org/10.1098/rspb.2014.1973>

Duguid, S., Wyman, E., Grueneisen, S., & Tomasello, M. (2020). The strategies used by

chimpanzees (*Pan troglodytes*) and children (*Homo sapiens*) to solve a simple

coordination problem. *Journal of Comparative Psychology*, 134(4), 401–

411. <https://doi.org/10.1037/com0000220>

Dunham, Y. (2018). Mere membership. *Trends in Cognitive Sciences*, 22(9), 780–

793. <https://doi.org/10.1016/j.tics.2018.06.004>

Eckerman, C. O., Davis, C. C., & Didow, S. M. (1989). Toddlers' emerging ways of achieving

social coordinations with a peer. *Child Development*, 60(2), 440–453.

Eckerman, C. O., & Didow, S. M. (1996). Nonverbal imitation and toddlers' mastery of verbal

means of achieving coordinated action. *Developmental Psychology*, 32(1), 141–

152. <https://doi.org/10.1037/0012-1649.32.1.141>

Eckerman, C. O., & Peterman, K. (2001). Peers and infant social/communicative

development. In G. Bremner & A. Fogel (Eds.), *Blackwell handbook of infant*

development (pp. 326–350). Blackwell Publishing.

Gelman, A., Carlin, J.B., Stern, H.S., Dunson, D.B., Vehtari, A., & Rubin, D.B. (2013).

Bayesian Data Analysis (3rd ed.). Chapman and Hall/CRC.

<https://doi.org/10.1201/b16018>

- Gelman, A., Simpson, D., & Betancourt, M. (2017). The Prior Can Often Only Be Understood in the Context of the Likelihood. *Entropy*, 19(10), 555.
<https://doi.org/10.3390/e19100555>
- Grüneisen, S., Wyman, E., & Tomasello, M. (2015). Children use salience to solve coordination problems. *Developmental Science*, 18(3), 495–501.
<https://doi.org/10.1111/desc.12224>
- Hrdy, S. (2006). Evolutionary context of human development: The cooperative breeding model. In: Dahlem Workshop No. 92., C. Carter & L. Ahnert (Eds), Attachment and Bonding: A New Synthesis (pp. 19– 46). Cambridge, MA: MIT Press.
- Howes, C., & Matheson, C. (1992). Sequences in the Development of Competent Play With Peers: Social and Social Pretend Play. *Developmental Psychology*, 28, 961-974. <https://doi.org/10.1037/0012-1649.28.5.961>.
- Köymen, B., Green, Y., Bennett, S., & Tomasello, M. (2026). Three-year-old children understand the false beliefs of their partner in collaborative decision making. *Child Development*, 97(1), 1–8. <https://doi.org/10.1093/chidev/aacaf001>
- Moreland, Richard. (2010). Are Dyads Really Groups? *Small Group Research*. 41. 251-267.
10.1177/1046496409358618.
- Ninio A. (2016). Bids for joint attention by parent-child dyads and by dyads of young peers in interaction. *Journal of Child Language*, 43(1), 135–156.
<https://doi.org/10.1017/S0305000915000082>
- Noyes, A., Keil, F. C., & Dunham, Y. (2018). The emerging causal understanding of institutional objects. *Cognition*, 170, 83–87.
<https://doi.org/10.1016/j.cognition.2017.09.008>

- Rubin, K. H., Bukowski, W., & Parker, J. G. (2006). Peer interactions, relationships, and groups. *Handbook of child psychology*, 3(5), 619-700.
- Schelling, T. C. (1960). *The Strategy of Conflict*. Harvard University Press.
- Tomasello, M. (2018). The normative turn in early moral development. *Human Development*, 61(4-5), 248–263. <https://doi.org/10.1159/000492802>
- Tomasello, M. (2019). *Becoming human: A theory of ontogeny*. Harvard University Press.
- Tomasello, M. (2024). *Agency and Cognitive Development*. Oxford University Press.
- Vanpaemel, W. (2010). Prior sensitivity in theory testing: An apologia for the Bayes factor. *Journal of Mathematical Psychology*, 54(6), 491–498. <https://doi.org/10.1016/j.jmp.2010.07.003>
- Warneken, F., Chen, F., & Tomasello, M. (2006). Cooperative Activities in Young Children and Chimpanzees. *Child Development*, 77(3), 640–663. <https://doi.org/10.1111/j.1467-8624.2006.00895.x>
- Warneken, F., Steinwender, J., Hamann, K., & Tomasello, M. (2014). Young children's planning in a collaborative problem-solving task. *Cognitive Development*, 31, 48–58. <https://doi.org/10.1016/j.cogdev.2014.02.003>

TABLES:

Age	Group Size	N (groups per trial)	Number of Successful Groups			Success Rate (% of test trials)
			Trial 1	Trial 2	Trial 3	
2-year-old	Dyads	10	8	9	8	83.3%
	Triads	10	4	5	5	46.7%
3-year-old	Dyads	10	10	10	9	96.7%
	Triads	10	9	9	9	90.0%

Table 1. Number of successful groups by age, group size, and trial.

	Dyads	Triads
2-year-olds	10	1
3-year-olds	10	10

Table 2. Number of successful groups in the final domain generalization trial.

Age	Group Size	Percentage of Trials (Test and Generalization)		
		Collaborative Success	Individualistic Success	Unsuccessful
2-year-old	Dyads	27.5%	60.0%	12.5%
	Triads	12.5%	35.0%	62.5%
3-year-old	Dyads	72.5%	25%	2.5%
	Triads	55.0%	37.5%	7.5%

Table 3. Type of strategy used by successful groups by age and group size.

FIGURE CAPTIONS:

Figure 1. Pure coordination task in triad condition.

Figure 2. Predicted Success Rates by Condition. Solid lines represent predicted success rates across all four trials by group age and group size (posterior means). Group age is the mean age of group members in years. Bands represent middle 95% posterior intervals. The y-axis represents the predicted probability of successfully opening the box on a given trial, with a probability of 0.5 representing chance (denoted by the horizontal broken line). The vertical broken line represents the youngest age at which dyads and triads succeeded at nondifferent rates (details below and in Supplementary Material Figure 2).

Figure 3. Communication Predicts Success Rates. Success rate predicted by groups' communication. Solid line represents posterior mean. Bands represent middle 95% posterior intervals.

Figure 4. Coordination Strategies. Values above .5 (i.e., chance, denoted by a broken horizontal line) indicate a tendency to use a collaborative strategy (dark purple), and values below a tendency to use an individualistic strategy (light purple). Percentages indicate relative evidence for the statistical hypothesis that collaborative strategies were used more often than individualistic strategies.

Figure 5. Key Color Selection Strategies. The x-axis represents the number of successful groups who chose keys according to the strategy depicted on the y-axis.